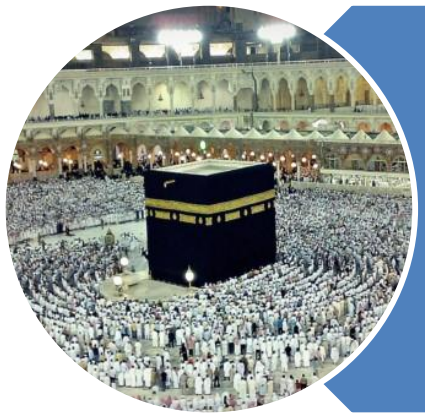
Data Structures - I

CPCS 204

رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

QUEUES

Introduction

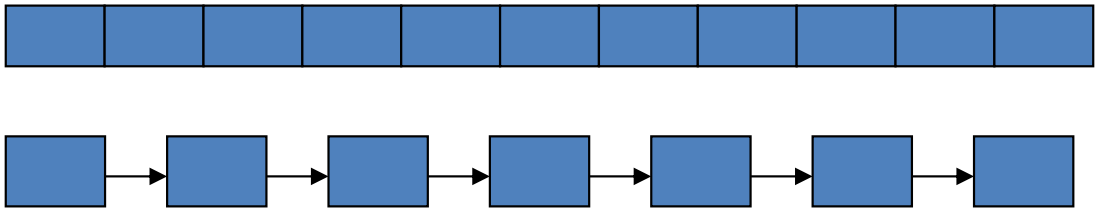
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

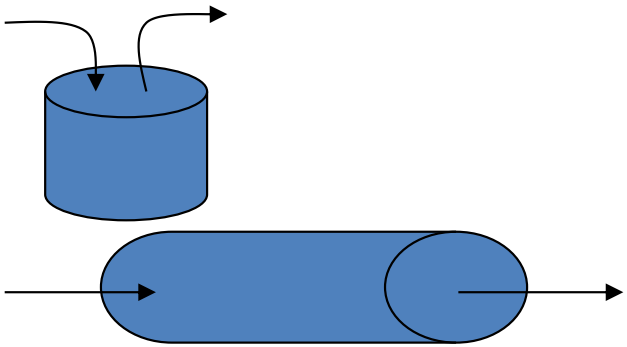
CPCS 204

Queues - Introduction

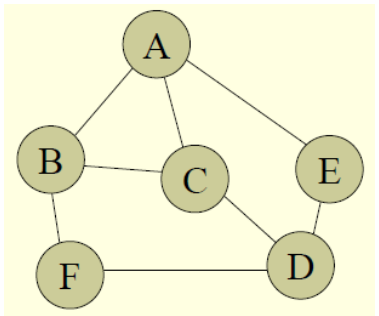
- Arrays
- Linked List

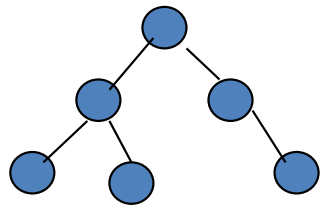


- Stacks
- Queues



- Tree
- Graph





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Queues - Introduction

- Definition
  - A data structure that stores information in the form of a typical waiting line
  - Placing a value one after the other
- Example
  - Queue in Al-Baik
  - Queue at Airport
  - Queue at Bus Stop

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CPCS 204

Queues - Introduction

- Add to the rear of queue
- Remove from front
- Abstract Data Type (ADT)
- Logical data structure
- Implementation
  - Arrays
  - Linked List

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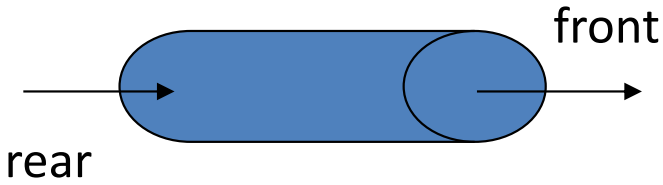
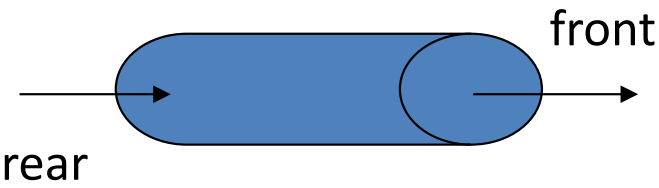
Queues - Introduction

- **Access Policy**
  - The first element that is inserted into the queue is the first element that will leave the queue
- **First In First Out**
  - FIFO
- **Processing End**
  - Front (to remove)
  - Rear (to add)



CPCS 204

Queues – Summary



- Abstract Data Type (ADT)
- Logical Data Structure
- Linear Data Structure
- Homogeneous
- Accessible only from both ends i. e. front & rear
- Access policy is FIFO
- Implemented using Arrays or Linked List

CPCS 204

QUEUES

Basic Operations & Its Use

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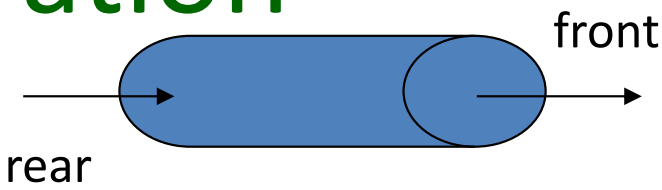
Dr. Muhammad Umair Ramzan

CPCS 204

Queues – Basic Operation



- enqueue
- dequeue
- peek
- isEmpty
- isFull
- clear

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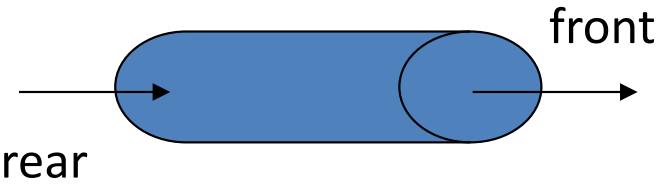
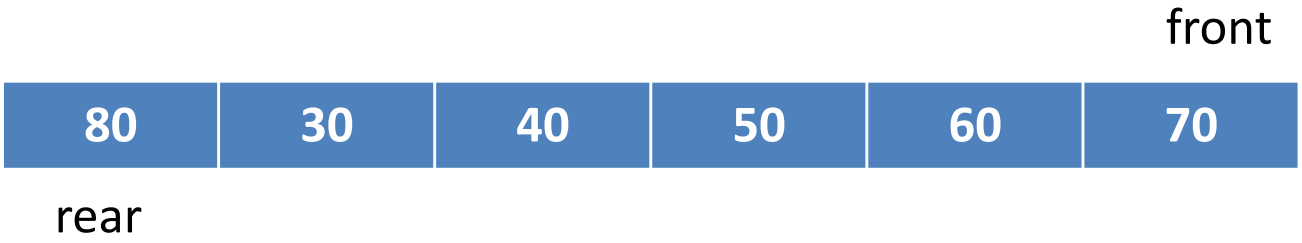
Queues – Basic Operation



- **enqueue**
  - Inserts an element at the end of the queue
  - **O(1)**

- **enqueue (80)**



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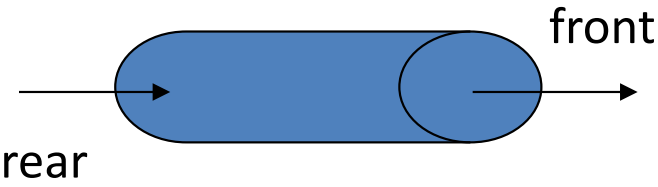
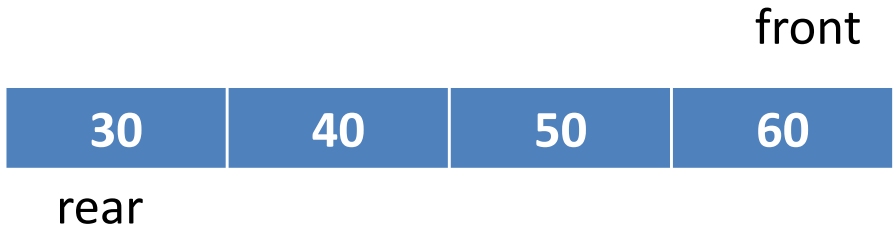
Queues – Basic Operation



- **dequeue**
  - Removes the element from the front of the queue
  - **O(1)**

- **dequeue ()**

Value = 70



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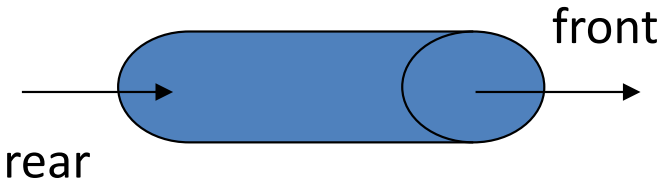
Queues – Basic Operation

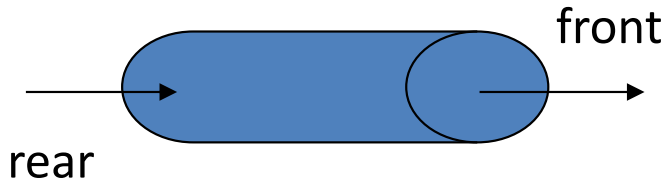


- **peek**
  - Return the element from the front of the queue without removing it
  - **O(1)**

- **peek ()**

  Value = 70

Queues – Basic Operation

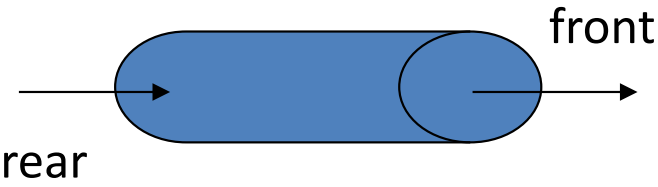


- **isEmpty**
  - Checks to see if the queue is empty
  - **O(1)**

- **isEmpty ()**

count = 0	**True**
otherwise	**False**

Queues – Basic Operation



- **isFull**
  - Checks to see if the queue is full
  - **O(1)**

- **isFull ()**

count = maxSize	**True**
otherwise	**False**

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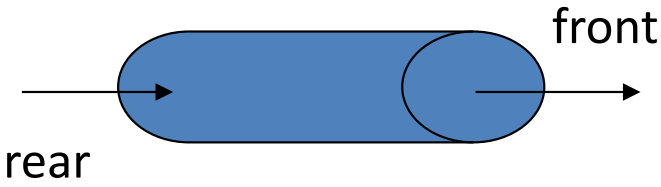
Queues – Basic Operation



- **clear**
  - Remove all values from the queue
  - **O(n)**

- **clear ()**

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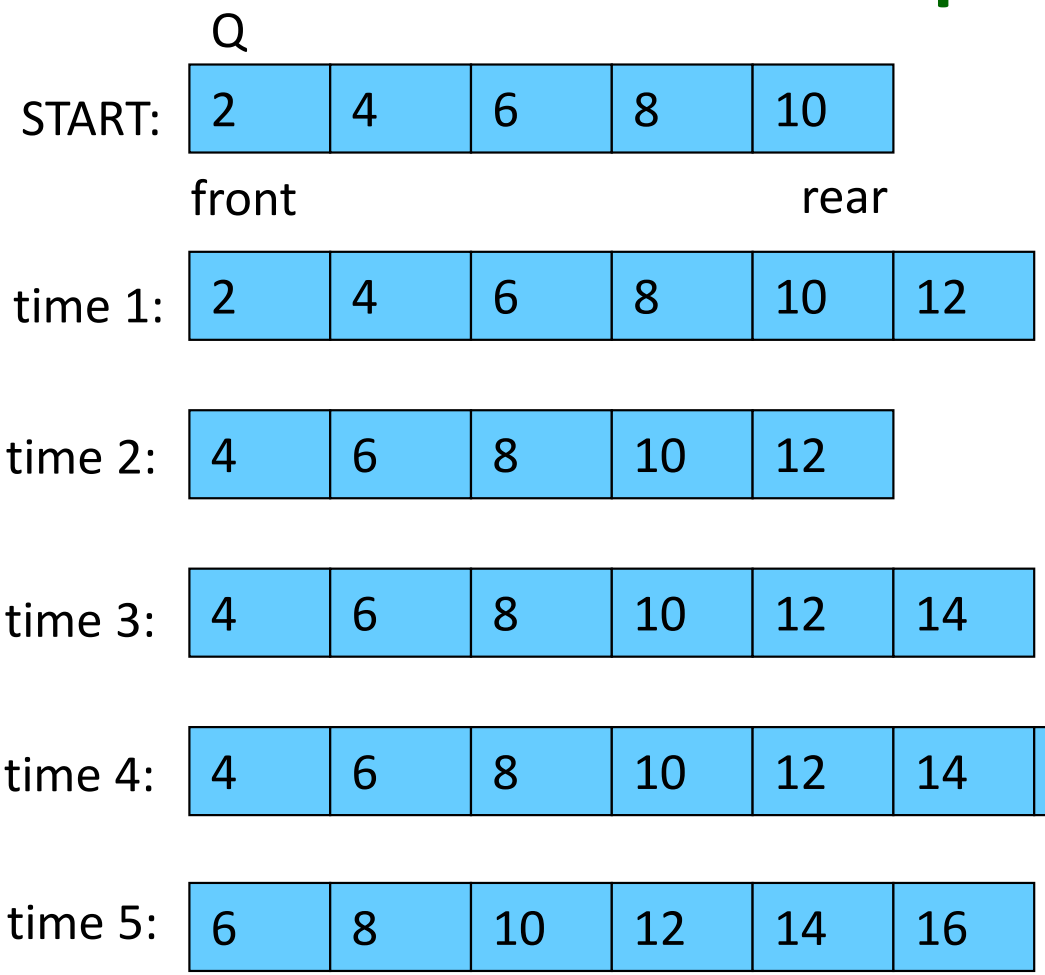
Queues – Basic Operation



Sequence of operations

Time	Operation
1	enqueue(12)
2	dequeue
3	enqueue(14)
4	enqueue(16)
5	dequeue

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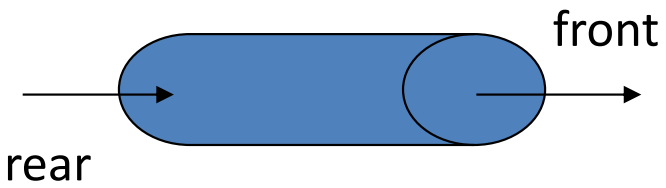
Queues – Basic Operation

- **enqueue** Modify the content
- **dequeue** Modify the content
- **peek** Do not modify
- **isEmpty** Do not modify
- **isfull** Do not modify
- **clear** Modify the content



QUEUES

Implementation Logic

"Brute Force" method

Front fixed & Rear moveable

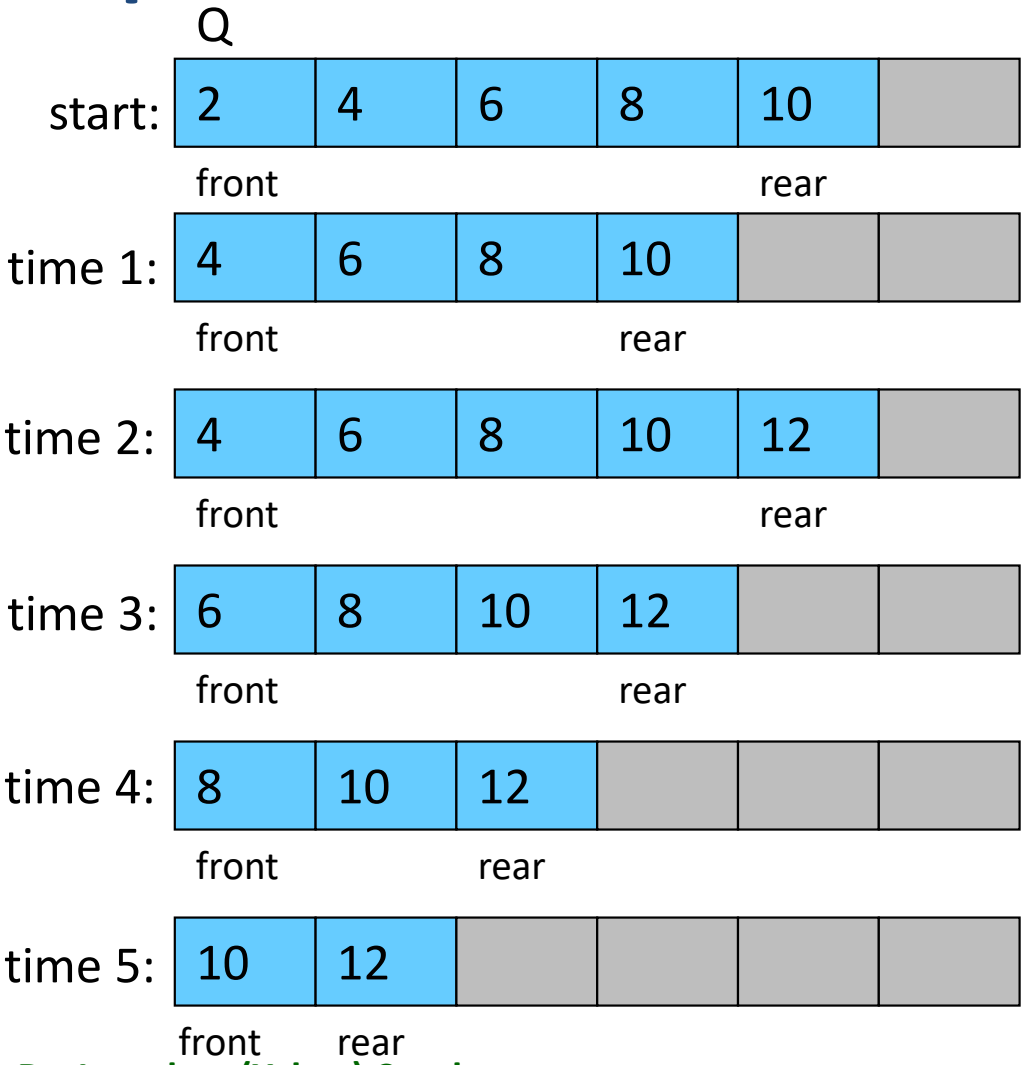
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Implementation – "Brute Force" method

- Real Time approach
- Service desk doesn't move
  - After customer at front served
  - He will leave the queue
  - All customers have to take one step ahead

Implementation – "Brute Force" method



Sequence of operations

Time	Operation
1	dequeue
2	enqueue(12)
3	dequeue
4	dequeue
5	dequeue
6	dequeue
7	dequeue

Notice that the array now has room to add elements to the queue.

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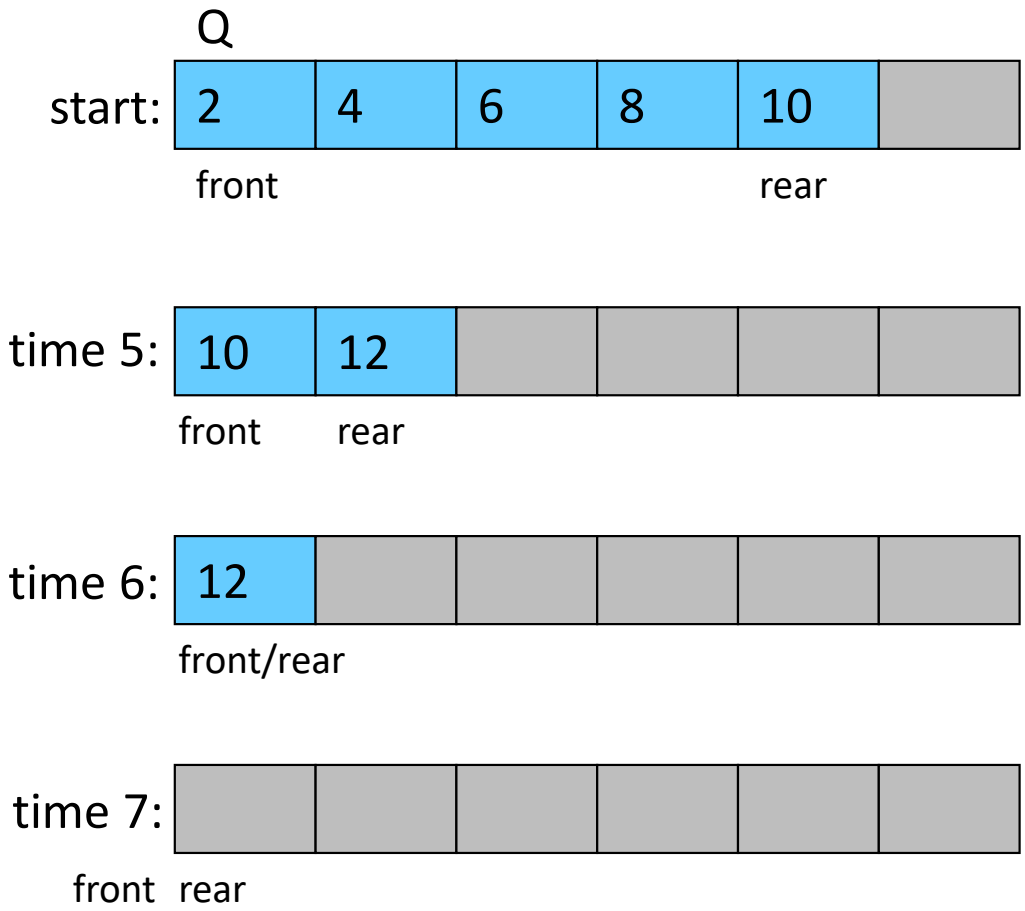
Implementation – "Brute Force" method



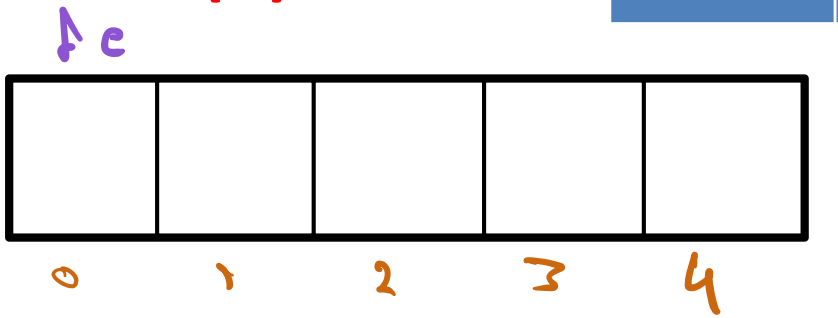
Sequence of operations

Time	Operation
1	dequeue
2	enqueue(12)
3	dequeue
4	dequeue
5	dequeue
6	dequeue
7	dequeue

Implementation – "Brute Force" method

- Insertion will be at rear
  - No shifting required
  - O(1)

- Removal will be from front
  - shifting required
  - **O(n)**



QUEUES

Implementation Logic

"Unrealistic" method

Front & Rear moveable

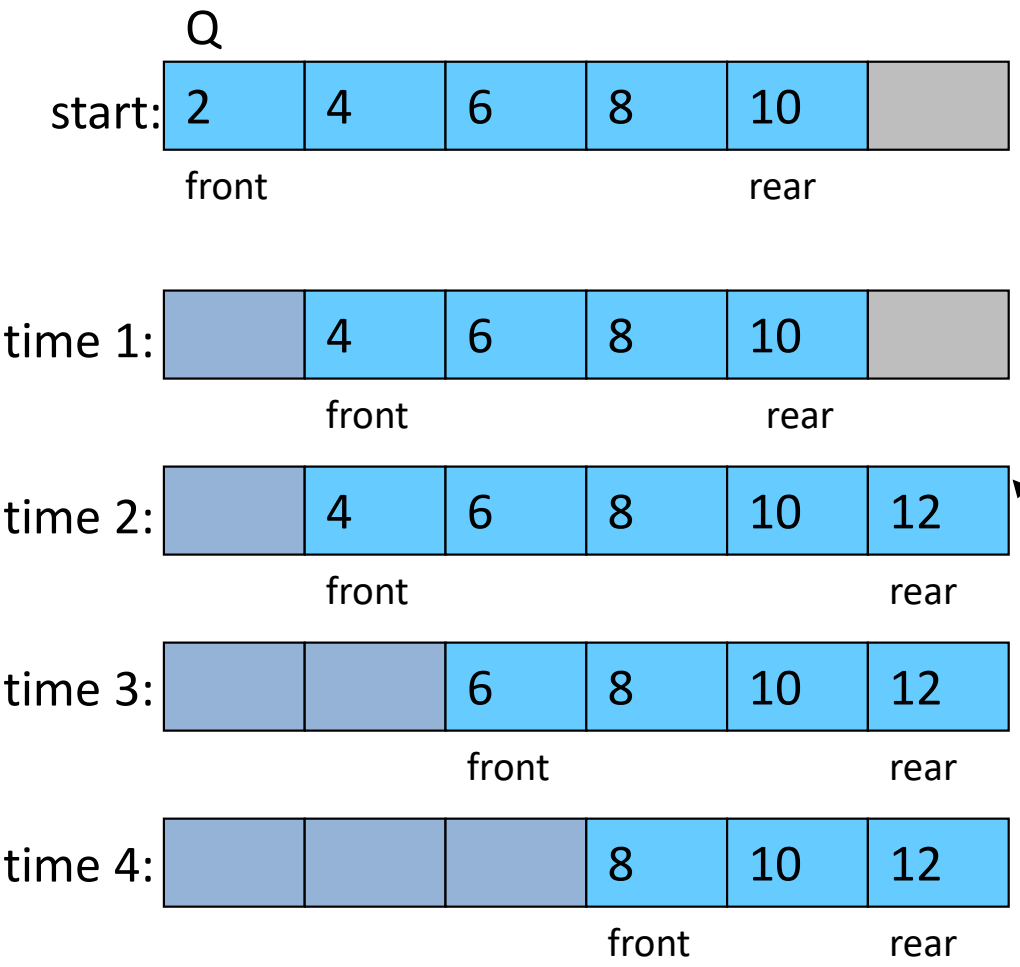
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Implementation – "Unrealistic" method

- Unrealistic approach
- Make service desk moveable
  - After customer at front served
  - Service desk will move to next person
  - No need to move the customers

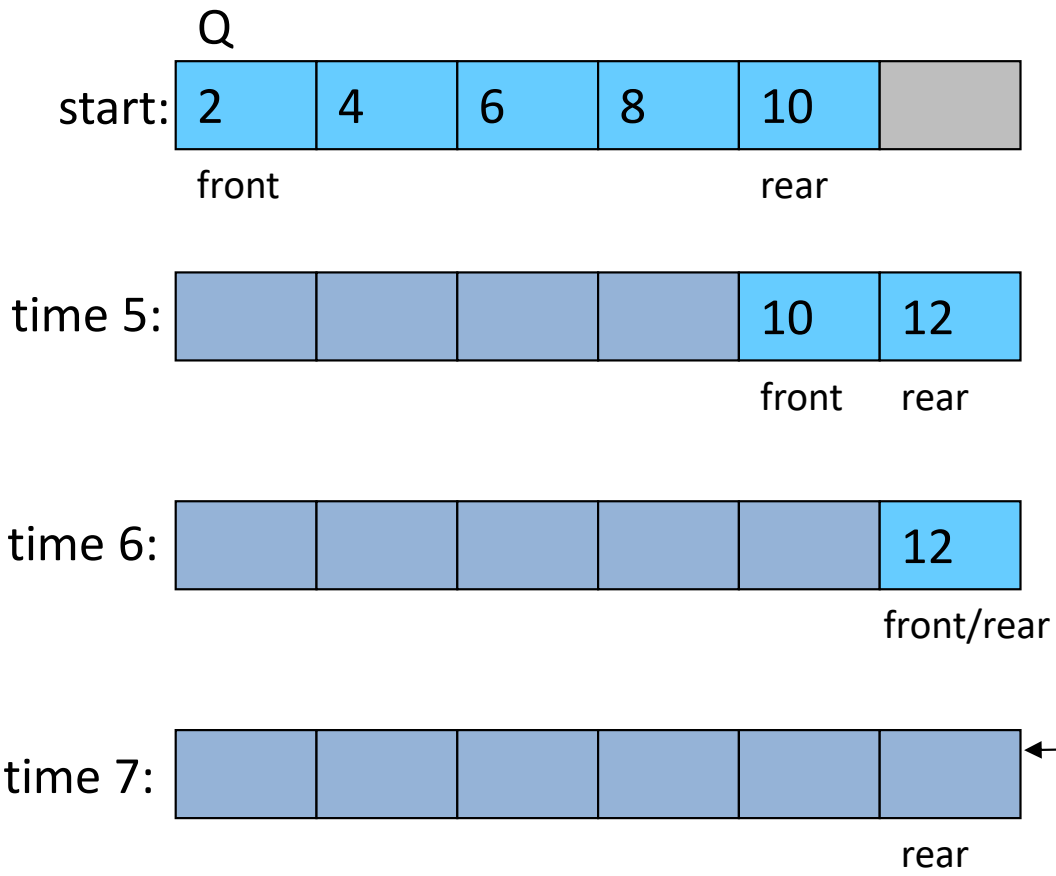
Implementation – "Unrealistic" method



Sequence of operations	
Time	Operation
1	dequeue
2	enqueue(12)
3	dequeue
4	dequeue
5	dequeue
6	dequeue
7	dequeue

Notice that the queue now appears to be full even though there are locations available in the array!

Implementation – "Unrealistic" method



Sequence of operations

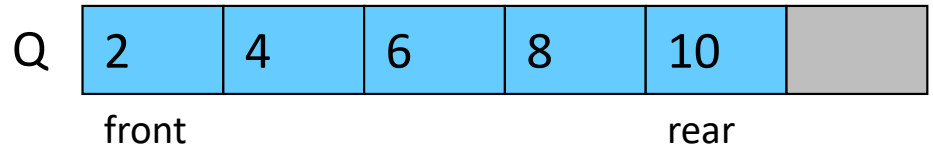
Time	Operation
1	dequeue
2	enqueue(12)
3	dequeue
4	dequeue
5	dequeue
6	dequeue
7	dequeue

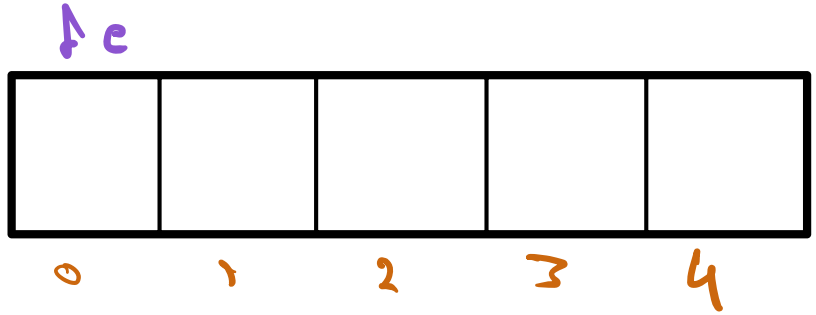
Even Worse:
The queue now appears full even though there are NO items in the array!

CPCS 204

Implementation – "Unrealistic" method

- Insertion will be at rear
  - No shifting required
  - Increment in rear
  - O(1)
- Removal will be from front
  - No shifting required
  - Increment in front
  - O(1)
- Problem
  - Space utilization is not optimal
  - Wasted cell





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QUEUES

Implementation Logic

Circular Arrays

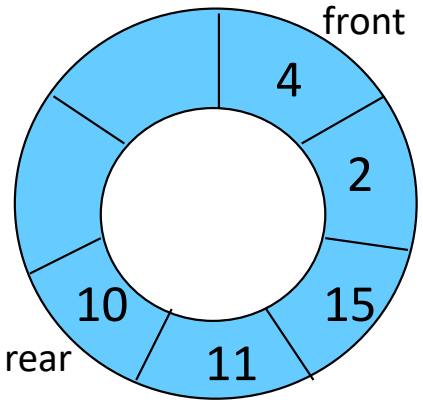
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Implementation – using "Circular Array"

- Common way of implementing an array-based queue
- It is a regular array

  - Increments will be made circular
  - It will behave like circular array
- Queue Empty
  - count = 0
- Queue Full
  - count = maxsize



visualization of circular array

Dr. Jonathan (Yahya) Cazalas

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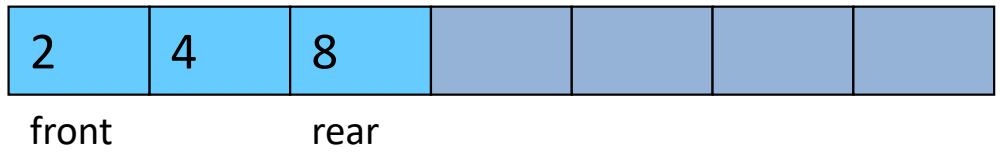
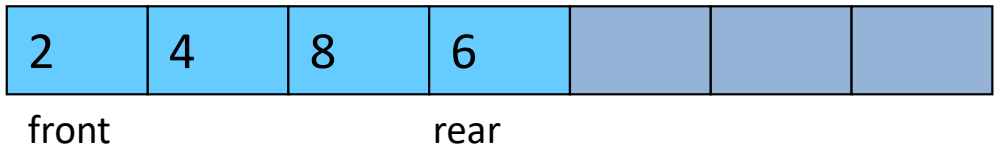
Implementation – using "Circular Array"

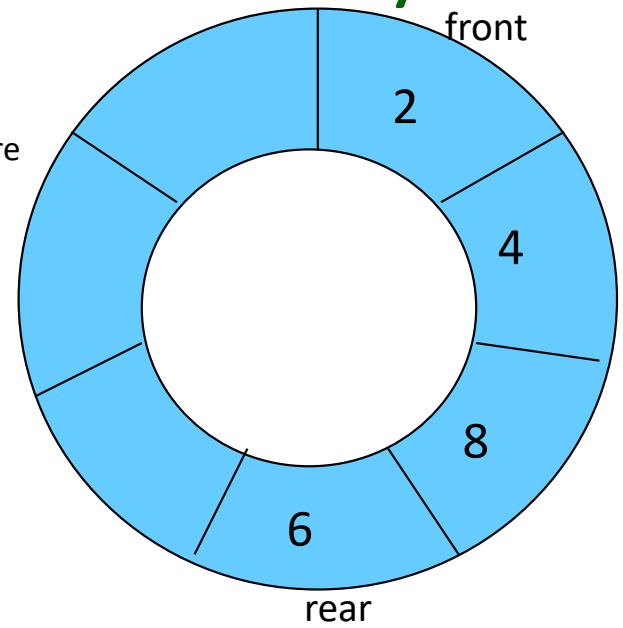


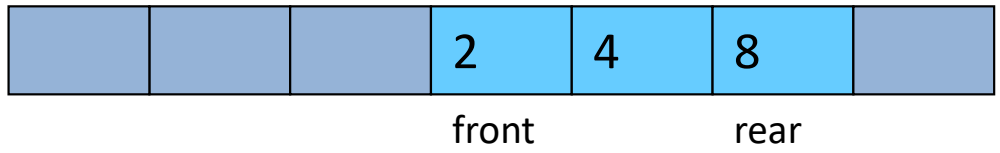
before



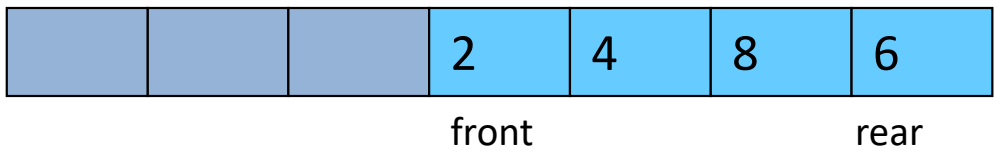
after

normal array implementation





before



after

circular array implementation

visualization of a queue implemented as a circular array after insertion of element 6

Dr. Jonathan (Yahya) Cazalas

Enqueue element 6

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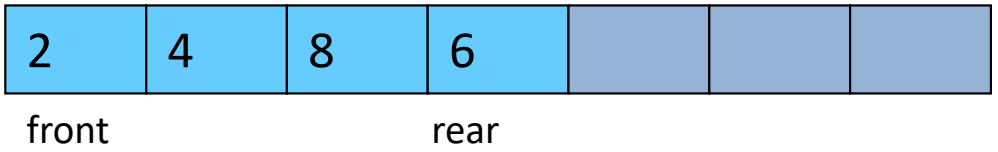
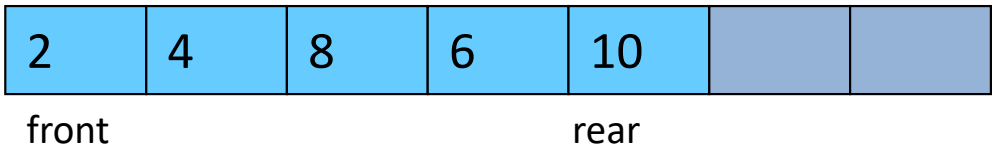
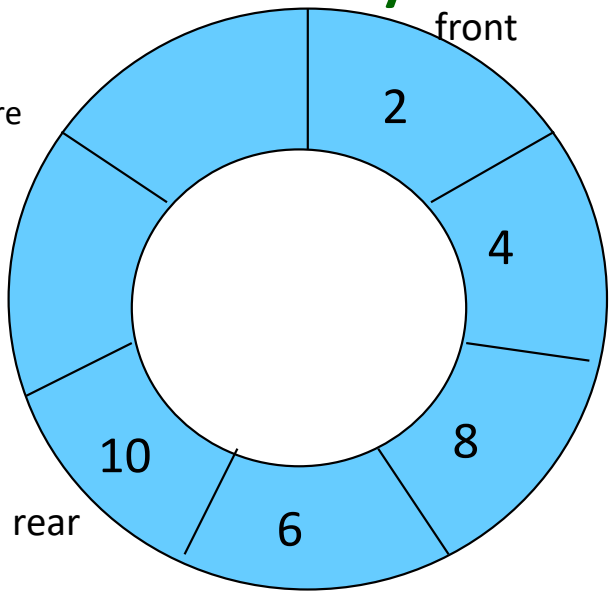
Implementation – using "Circular Array"

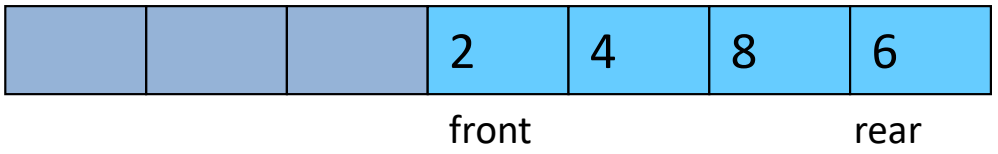


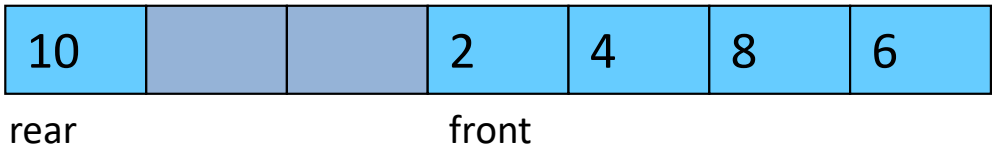


normal array implementation



visualization of a queue implemented as a circular array after insertion of element 10

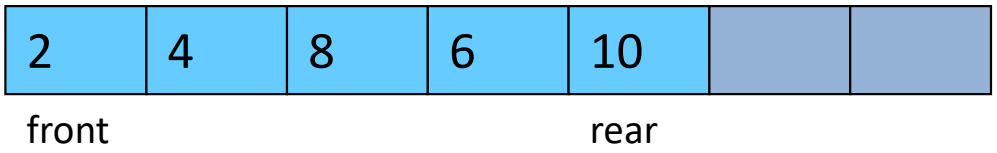




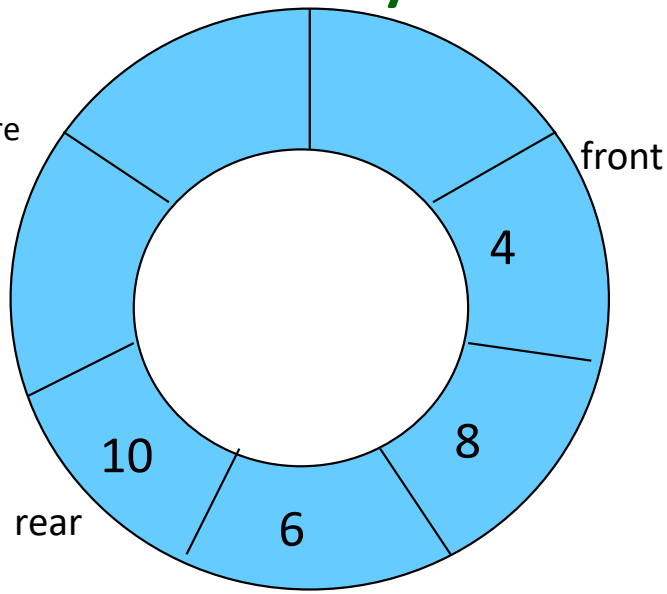
circular array implementation

CPCS 204

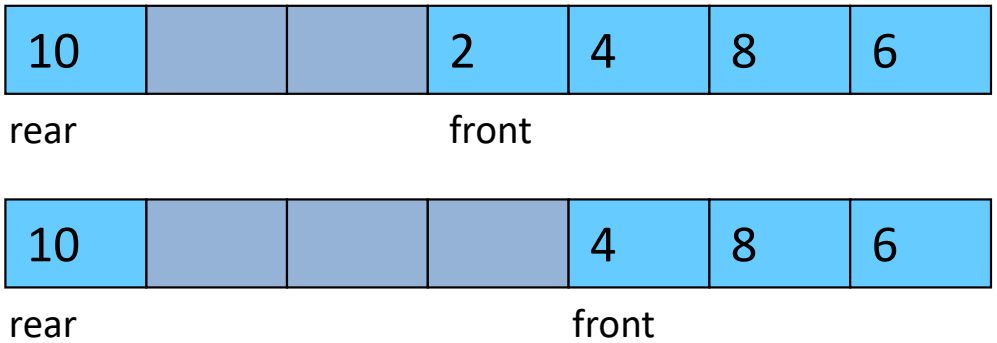
Implementation – using "Circular Array"



normal array implementation



visualization of a queue implemented as a circular array after dequeue operation



circular array implementation

dequeue

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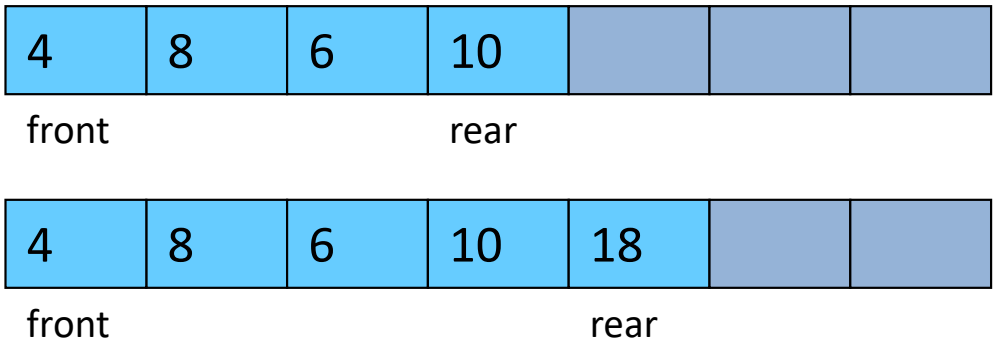
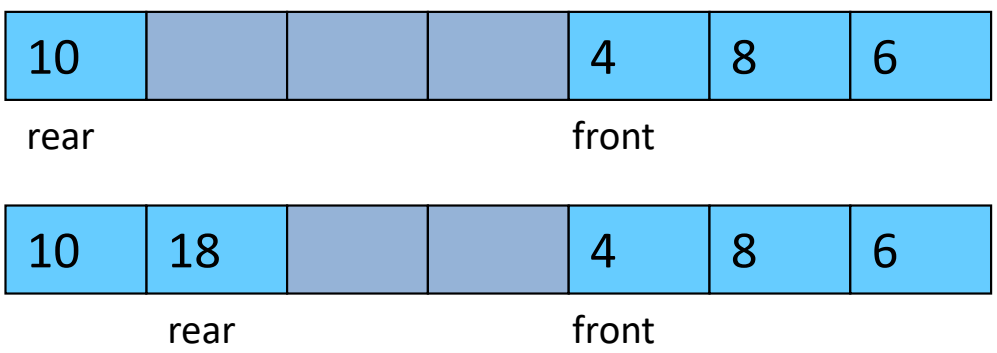
CPCS 204

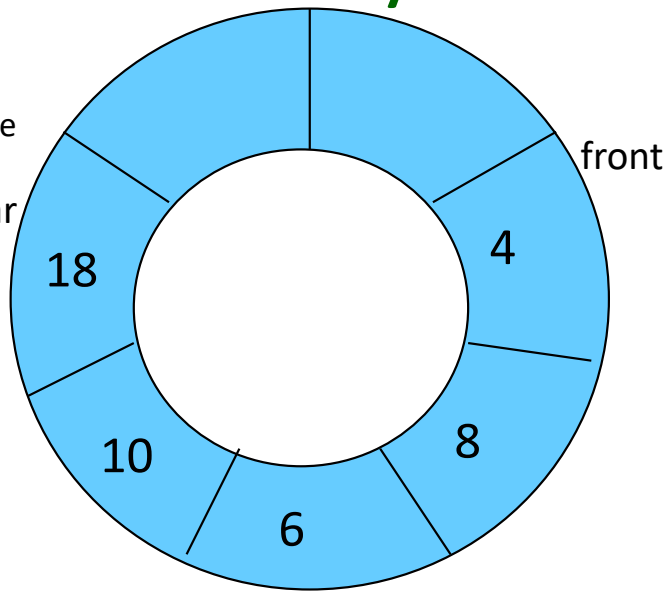
Implementation – using "Circular Array"



normal array implementation



circular array implementation



visualization of a queue implemented as a circular array after insertion of element 18

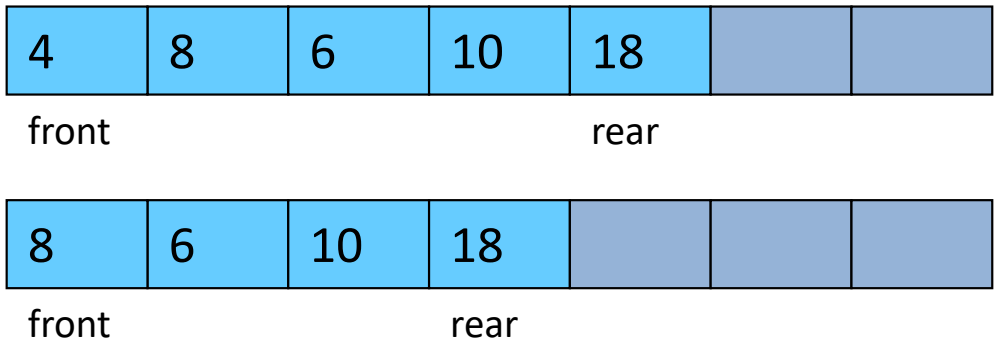
Dr. Jonathan (Yahya) Cazalas

Enqueue element 18

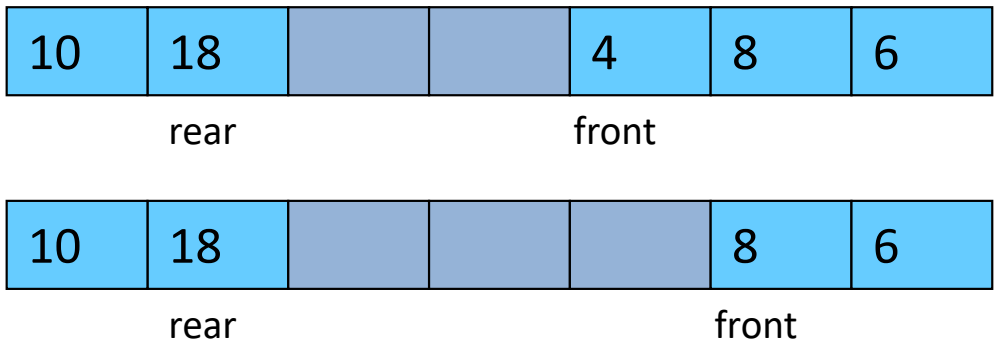
Dr. Muhammad Umair Ramzan

CPCS 204

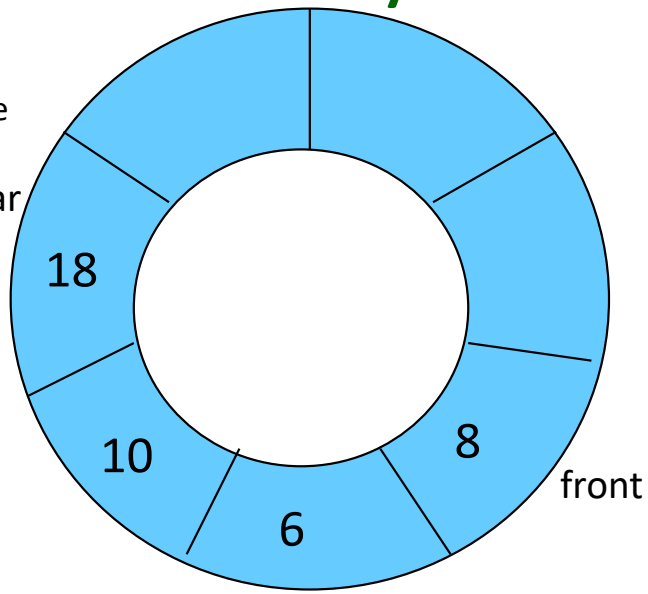
Implementation – using "Circular Array"



normal array implementation

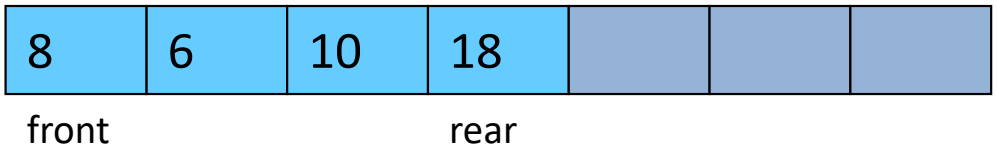


circular array implementation

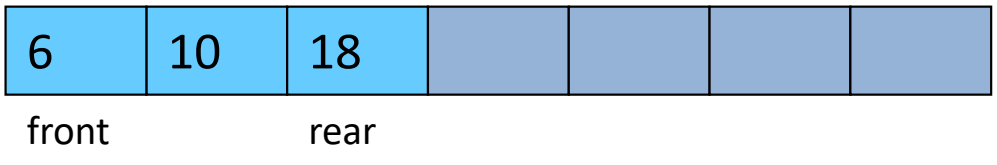


visualization of a queue implemented as a circular array after dequeue operation

Implementation – using "Circular Array"

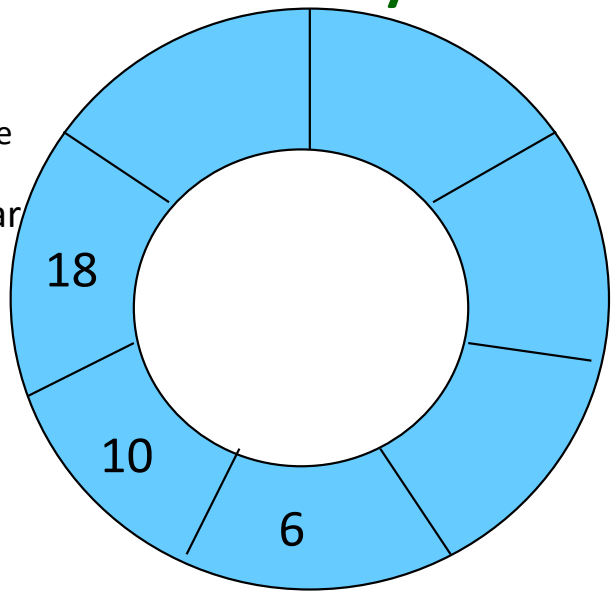


before

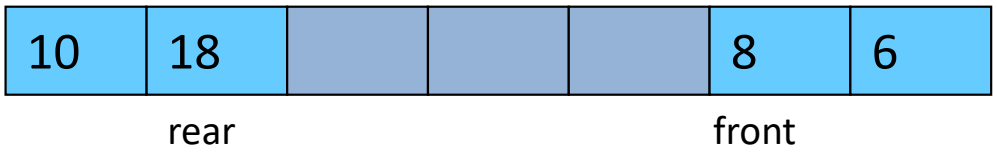


after

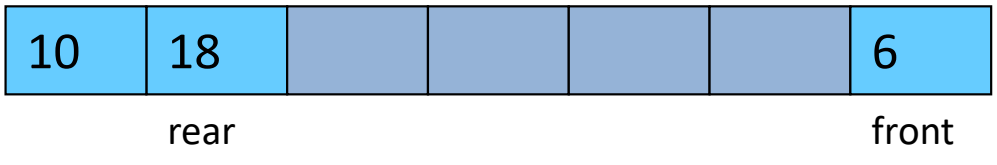
normal array implementation



visualization of a queue implemented as a circular array after dequeue operation



before

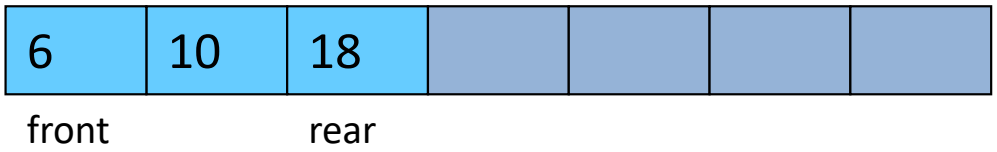


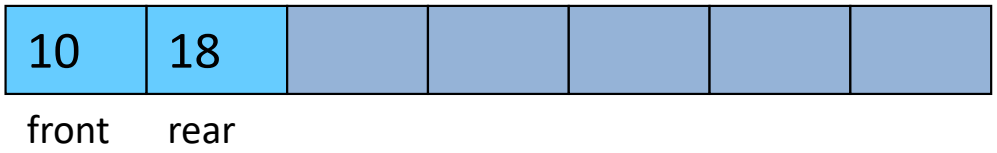
after

circular array implementation

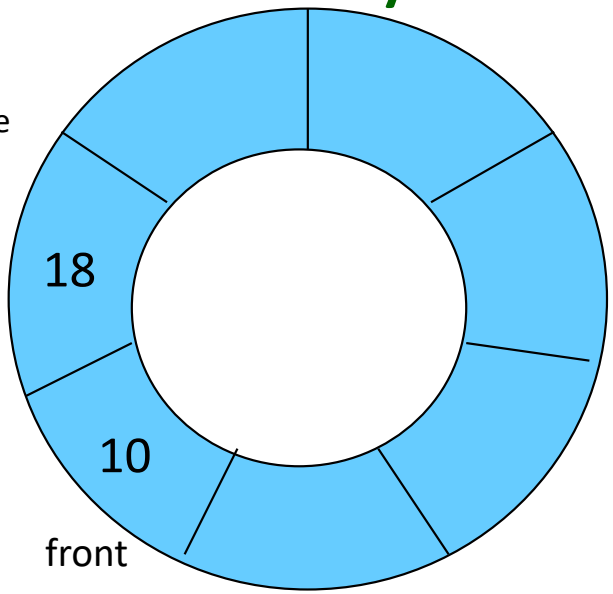
CPCS 204

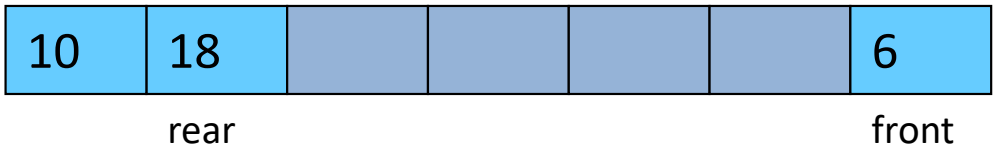
Implementation – using "Circular Array"

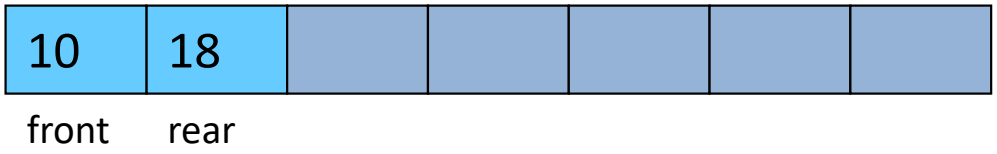




normal array implementation







circular array implementation

visualization of a queue implemented as a circular array after dequeue operation

dequeue

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

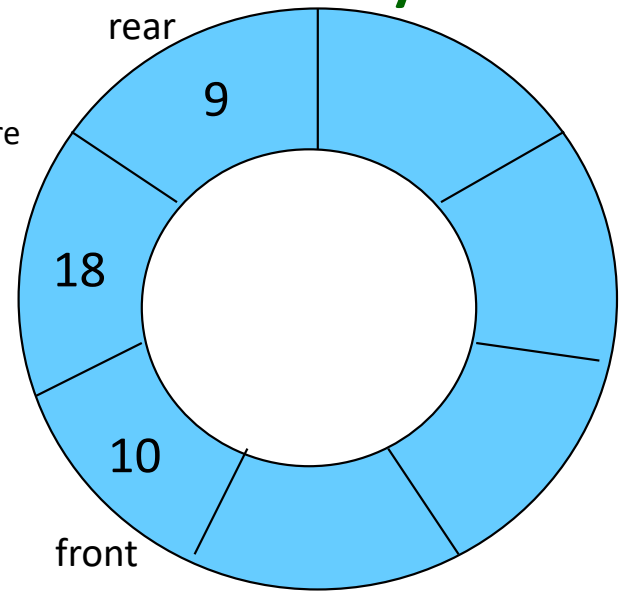
CPCS 204

Implementation – using "Circular Array"

before

after

normal array implementation



visualization of a queue implemented as a circular array after insertion of element 9

before

after

circular array implementation

Dr. Jonathan (Yahya) Cazalas

Enqueue element 9

Dr. Muhammad Umair Ramzan

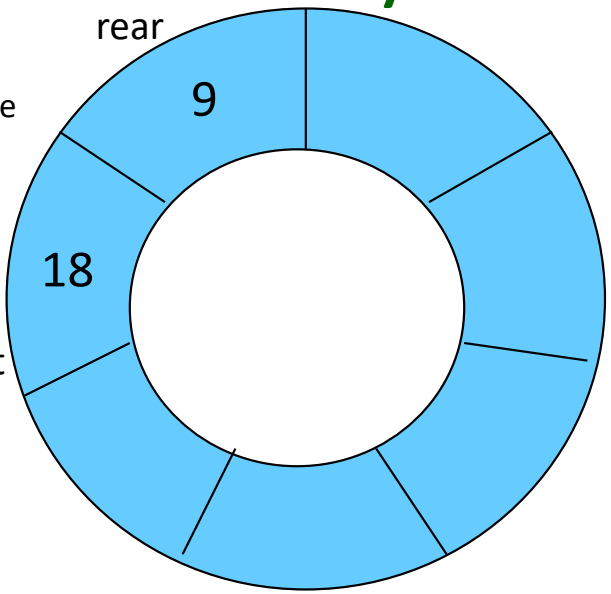
CPCS 204

Implementation – using "Circular Array"

before

after

normal array implementation



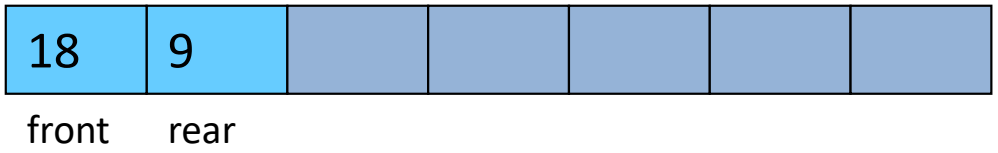
visualization of a queue implemented as a circular array after dequeue operation

before

after

circular array implementation

Implementation – using "Circular Array"



before

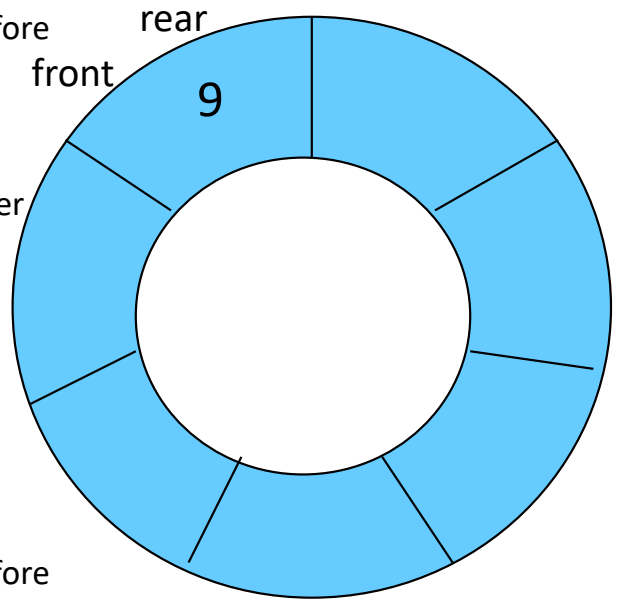
after

normal array implementation

before

after

circular array implementation



visualization of a queue implemented as a circular array after dequeue operation

dequeue

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

CPCS 204

Implementation – using "Circular Array"

before

front/rear

after

front/rear

normal array implementation

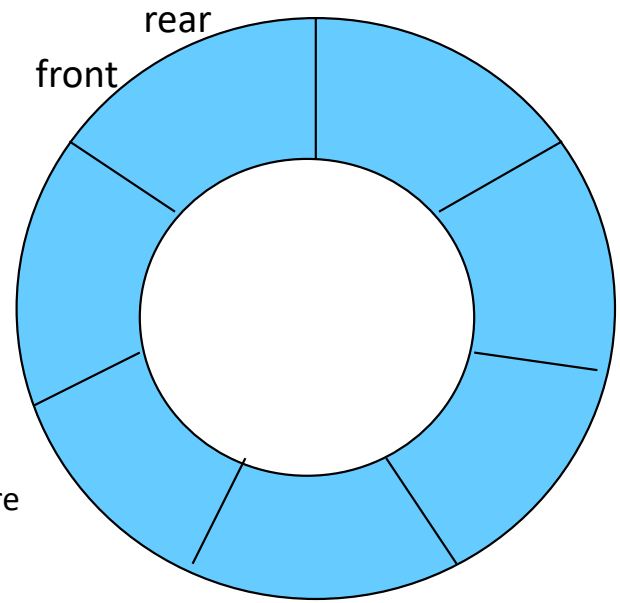
before

front/rear

after

front/rear

circular array implementation



visualization of a queue implemented as a circular array after dequeue operation

dequeue

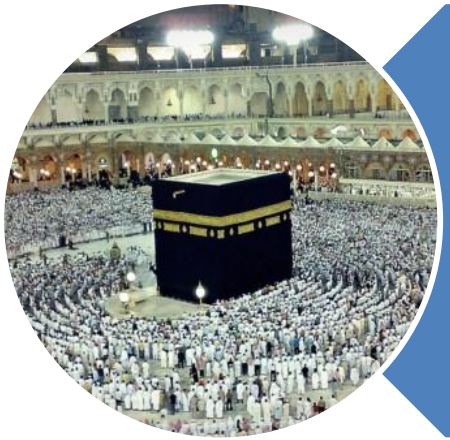
Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan

Data Structures - I

"No Bonus" marks for this course anymore

CPCS 204



Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us).(Quran 3 : 173)

Dr. Jonathan (Yahya) Cazalas

Dr. Muhammad Umair Ramzan